

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CSC 202 — Past Questions & Materials
by Renance

Course: CSC 202
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

This study pack was compiled and restructured by Renance for CSC 202 students at Federal University of Petroleum Resources, Effurun. It brings together 3 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 3 — CSC 202 - 2024/2025 - Second Semester - COMPUTER PROGRAMMING II

Level 200 • Second Semester • PastQuestion

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\D SEMESTER EXAMINATION20242025 SESSION Soa Ce

OE TEPLE: COMPUTER PROGRAMMING {1 COURSE CODE: CSeqp7

SUCTION: ANSWER QUESTION ONE AND ANY 3 OVHp\$ QUESTIONS Time: 2HRS1S PF

; a NS

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PION ONE . a - ee

) Write a program that will determife the addition

income by subtracting a M5000 allowance for

number of dependent, and tax a:ndunts ahead

between tax owed and (ax deducted followed

this difference is positive or negative, Imks

al state tax ped by an employee. If the State

each dependent from gross income. Your pro

charges 4.4% tax on net

gram should read gross inco

Y deducted, I Wifi icn Compute the actual tay owed and print tic diffe

by the message «ND CHEQUE” or “Re ry» depending on whether

') Write a program that solves i the real roots of the quadratic Equation $ax^2 + bx + e = 0$, Using the formula $x = \frac{-b \pm \sqrt{b^2 - 4ae}}{2a}$

. Note: $b^2 - 4ae$ is the discriminant. 8mks ta

) A leap year is any year divisible by 4 unless it is divisibl : h.

. eb but not 400. Write a -

leap year. \$mks ¥ 100, Program to tell whether a year isa

TION TWO

To convert temperatures written in Fahrenheit to Celsius ct 32, multiply by 5 and then divi

Celsius to Absolute value (kelvin), you add 2 » You subtract 32, mulliply OY > and'then divide by 9.

To convert

73.5. Write a ram that displays a temperaty ;

Peeeeh Garis prog play peas conversion chart on the

| There are 9870 people in Ugbrome town, whose population increases by 10% each year,

baag.to determine haw

> Write a C++ to find the engi of heen for the population to exceed 30,000. smi

rd side of a triangle, if we know the
 ; : : gic, now the length
 triangle and the angle between them in degrees (alpha), Use
 the formula $a^2 = b^2 + c^2 - 2bc \cos(\alpha)$
 » TION THREE

i
 ' | Your company pays its hourly workers once a week. An employee
 fee's pay is based upon the number of hours worked (to
 the nearest half hour) and the employee's hourly pay rate. Weekly hours exceeding 40 are paid at a
 rate of time and a half.

Employees who earn over \$100 a week must pay union dues of \$5 per week. Write a payroll program
 that will determine
 the gross pay and net pay for an employee. 6mks

2) Write a simple C++ program that computes both the summation and factorial of the numbers 1
 through 50. Declare the
 loop variable i inside the for loop. 4mks

| The Pythagorean Theorem states that the sum of the squares of the sides of a right triangle is
 equal to the square of the
 hypotenuse. Given two positive numbers, m and n, where $m > n$ and the formulas: $side1 = m^2 - n^2$,
 $side2 = 2mn$,
 $hypotenuse = \sqrt{side1^2 + side2^2}$. Write a program that reads in values for m and n and prints the
 values of the Pythagorean
 triple generated by the formulas. 5mks a. j

QUESTION FOUR i, y

) Using a do-while statement, write a program that computes factorial function: $n! = n(n-1)(n-2)\dots(2)(1)$.
 5mks .

, Write a program to find the nearest whole number of a given number. n is the largest integer whose
 square is less than or
 equal to the given number. 4mks / mark

» Write a program using a for statement that will print a list of first 10 perfect
 squares, their squares and square roots on a table. 5mks a

QUESTION FIVE ; |
 on of exams scores ranging from 0 to 100. Your program should count and find
 0), number of satisfactory scores (60-89) and the number of unsatisfactory scores (0-59)
 ; i

.) Write a program to read in a column of outstanding scores (90-100). It should also display the count of
 59), It should also display the count of

4} What is displayed by the segment

```
Sum = 0;
while (sum <= 100)
sumt=5;
cout <<su
?ewrite the loop so that it pri
| Write a simple prograin usin
Mirough 100 inclusive. 4mks
the sum of the squares of integr
```

Paper 2 of 3 — CSC202 test 2025

Level 200 • Second Semester • PastQuestion

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CSC202 (COMPUTER PROGRAMMING 11) CEST — SLT OPTION

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ED AG Devlare the loop variable dinside the Jor loop.

A lean your ic. an

P YOU Ss ANY Year divisible py d unless itis divisible by 160, but net 400. Write a progrant 0
tell Whethep a year is dieap year

Paper 3 of 3 — CSC 202 COMPUTER PROGRAMMING II 2025

Level 200 • Second Semester • PastQuestion

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SD SEMESTER CNAMINATION 20242028 SESST 7. j ao

hava ' t 2038 SESSION COURSE CODE: €

XE TULLE: COMPUTER PROGRAMMIN oo

AUCTION, ANSWER QUESTION ONE AND ANY 3 OTHER QUESTIONS Time: .2H&S1S MINS

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STTON ONE

2} Yorke 9 program thst will determing the additiona! state tax owed by an employee. If the State charges a 4% tax on net

sacame ty subtracting a M5600 allowances for each dependent fron gross income. Your program should read gross income,

semmbor of depondent, and tax amounts already deducted. It will then compute the actual tax owed and print the difference

noumnon man owed and tax deducted followed by the message "SEND CHEQUE" or "REFUND" depending on whether

his difForcace is positive or negative. Imks

'rte a program thal solves for the real roots of the quadratic equation $ax^2 + bx + c = 0$. Using the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

'aie: $b^2 - 4ac$ is the discriminant. 8mks

A weap year is any year divisible by 4 unless it is divisible by 100, but not 400. Write a program to tell whether a year is a

Soar wear. Smks

sTION THO

To convert temperatures written in Fahrenheit to Celsius, you subtract 32, multiply by 5 and then divide by 9. To convert

Celsius to Absolute value (kelvin), you add 273.5. Write a program that displays a temperature conversion chart on the

screen. Gamks

| There are 9870 people in Ugbromo town, whose population increases by 10% each year, write a program using a while

toop w determine how many years it will take for the population to exceed 30,000, Smks

write 2 C~ 10 find the length of the thir le of a triangle, if we know the length of the other two sides (6 and 6) of a

triangle and the angle between them in degrees (α). Use the formula $a^2 = b^2 + c^2 - 2bc \cos(\alpha)$. 4mks

+ 110N THREE

=} Your company pays its hourly workers once a week. An employee's pay is based upon the number of hours worked (to

pay rate, Weekly hours exceeding 40 are paid at a rate of time and a half.

the nearest half hour) and the employee's hourly

Employees who work over 100 a week must pay union dues of NZ\$ per week. Write a payroll program that will determine

gross pay and net pay for an employee, 6mks

0) Write a simple C++ program that computes both the summation and factorial of the numbers 1 through 50. Declare the

loop variable / inside the for loop. 4mks

c) The Pythagorean Theorem states that the sum of the squares of the sides of a right triangle is equal to the square of the

hypotenuse. Given two positive numbers, m and n , where $m > n$ and the formulas: $side1 = m^2 - n^2$, $side2 = 2mn$,

hypotenuse = $\sqrt{side1^2 + side2^2}$. Write a program that reads in values for m and n and prints the values of the Pythagorean

triple generated by the formulas. 5mks

. SECTION FOUR

a) Using a do-while statement, write a program that computes the factorial of a number n :

>) Write a program to find the integer square root of a number n :

equal to the given number. 4mks

c). Write a program using a for statement to compute the sum of the squares of the first 100 integers.

SECTION FIVE

factorial function: $n! = (n)(n-1) \dots 3(2)(1)$. Sinks

That is the largest integer whose square is less than or equal to n .

that will print a list of integers, their squares and square roots on a table. 5mks

marks scores ranging from 0 to 100. Your program should count and print the number of scores in each category.

ia) Write a program to read in a collection of exam marks and the number of unsatisfactory scores (0-59).

9) and the number of unsatisfactory scores (0-59).

number of outstanding scores (90-100), number of satisfactory scores (60-89).

It should also display the category of each score read in. 6mks

(b) What is displayed by the segment.

Sum = 0;

while (sum <= 100)

sum+=5;

cout << sum<<endl;

Rewrite the loop so that it prints all multiples of 5 from 0 through 100 inclusive. 4mks

fc) Write a simple program using a for and while loop to find the sum of the squares of integers from 1 to n. 5mks